

Total No. of Questions : 8]

SEAT No. :

PD-4106

[Total No. of Pages : 2

[6402]-66

S.E. (Artificial Intelligence and Machine Learning)

COMPUTER NETWORKS

(2019 Pattern) (Semester-III) (218543)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Attempt Q.1, or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.
- 5) Use of Calculator is allowed.

- Q1)** a) What is CRC polynomial? Describe in detail with example. [9]
b) What is frames in data link layer? Describe in detail the fixed size framing and variable size framing? [9]

OR

- Q2)** a) What is noisy protocols? Describe the two protocols for noisy channels? [9]
b) Explain CSMA/CA random access technique in detail with suitable diagram/ flowchart, Explain how collision is avoided in CSMA/CA. [9]

- Q3)** a) What is default routing? Describe in detail RIP and EIGRP. [9]
b) Explain following terms [8]
i) NAT
ii) DHCP

P.T.O.

OR

- Q4)** a) Explain following terms: [9]
i) Distance Vector Routing
ii) Link State Routing
iii) Path Vector Routing
b) Explain ARP protocol with diagram. [8]
- Q5)** a) What is socket? Enlist all socket primitives & explain any three. [9]
b) Explain the three-way handshake algorithm for TCP connection establishment. Compare and contrast between TCP and UDP. [9]

OR

- Q6)** a) Explain TCP with its header format. [9]
b) What do you mean by flow control in transport layer? List the different methods of achieve it. Explain any one method in detail. [9]
- Q7)** a) What is the purpose of DNS? How does recursive resolution differ from iterative resolution? Explain with suitable diagram. [9]
b) What is MIME? What is the purpose of MIME? Explain MIME header in detail. [8]

OR

- Q8)** a) Describe the functions of the two FTP connections. List difference between FTP & TFTP. [8]
b) Write short note on [9]
i) IMAP
ii) SMTP
iii) SNMP

